

HIGH ACRYLONITRILE NBR PATENT RESEARCH REPORT

1. ABSTRACT

This report provides a technical summary of the research scope and target compound High Acrylonitrile NBR, analyzing 1 primary patent. The major technical approaches observed include polymerization and synthesis methods, with key process trends identified in temperature, pressure, and conversion ranges. No competitor patents or website sources were analyzed.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

Patent Details - Patent Number: EP3902855 - Patent Title: Pulverulent mixture of precrosslinked nitrile rubbers - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2026 - Material Type: Nitrile rubber - Relevance to target: Relevant to High Acrylonitrile NBR

Polymerization Method - Temperature: 100 °C (Mooney viscosity ML 1+4 @100°C) - Viscosity: 10 Mooney units (MU) (Mooney viscosity ML 1+4 @100°C) - Conversion: 80 % (Polymerization conversion) - Pressure: 100 °C (Reduced pressure at temperatures < 100°C) - Nitrile: 100 parts by weight (Nitrile rubber) - Emulsifier: 0.1 % (Emulsifier amount) - Monomer: 100 parts by weight (Monomer mixture) - Based: 500 % (Molar amount of reducing agent) - Temperature: 10 °C (Latex coagulation temperature) - Temperature: 90 °C (Wash temperature) - Temperature: 150 °C (Drying temperature) - Through: 100 °C (Latex heating temperature) - Pressure: 500 bar (Spraying pressure) - Temperature: 350 °C (Hot air inlet temperature) - Temperature: 60 °C (Glass transition temperature) - Overall: 100 parts by weight (Total amount of rubber additives) - Rubber: 150 parts by weight (Rubber additives) - Iii: 36 % (Monomer conversion) - Dispersion: 50 % (Dispersion of Vulkanox BKF) - Rubber: 100 parts (Vulcanizable mixture) - Temperature: 100 °C (Measurement temperature) - Tan: 60 °C (Loss factor temperature) - Tan: 23 °C (Loss factor temperature) - Tan: 0 °C (Loss factor temperature) - Amplitude: 0.5 % (Amplitude sweep) - Amplitude: 15 % (Amplitude sweep) - Temperature: 8 bar (Ram pressure) - Fill: 70 °C (Fill level temperature) - Pressure: 8 bar (Ram pressure) - Mixing: 150 °C (Mixing temperature) - 0-210: 72 % (Fill level) - Temperature: 40 °C (Roll temperature) - Elongation: 15 % (Elongation)

Relevant Experimental Evidence - The nitrile rubbers have a Mooney viscosity ML 1+4 @100°C of 10 to 180 Mooney units (MU). - The polymerization temperature is typically in the range from

0°C to 50°C. - The conversion is conducted up to a conversion of at least 80%, based on the monomer mixture used. - The latex coagulation is performed in the temperature range of 10°C to 110°C. - The wash is conducted at a temperature in the range from 15°C to 90°C.

Technical Relevance This patent is relevant to High Acrylonitrile NBR as it provides a method for producing a pulverulent mixture of precrosslinked nitrile rubbers, which can be used in various applications.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

Trend Analysis: - Insufficient comparable evidence was available to identify a reliable cross-patent trend.

4. CONCLUSION

The analyzed patent provides a method for producing a pulverulent mixture of precrosslinked nitrile rubbers, which can be used in various applications. The technical parameters and conditions for the production process are well-defined, providing a clear understanding of the requirements for producing High Acrylonitrile NBR.

5. REFERENCES

- [EP3902855](#) - Pulverulent mixture of precrosslinked nitrile rubbers